BEHAVIOURAL FINANCE MINI-REPORT

A Study of Price Movements and Investor Behaviour: Hindustan Aeronautics Limited (HAL)

Subject: Behavioural Finance
Company Selected: Hindustan Aeronautics Limited (HAL)
Data Period: 14 October 2025 – 14 October 2026
Market Benchmark: NIFTY 50

INTRODUCTION

This report analyses the daily share-price movements of Hindustan Aeronautics Limited (HAL), an Indian aerospace and defence company. The study examines daily closing prices, daily returns, trading volumes and major company announcements. HAL's share-price movements are also compared with the NIFTY 50 index to understand whether the company's performance differed from the broader market.

The report also examines HAL's official clarification regarding the Advanced Medium Combat Aircraft (AMCA) programme, released on 4 February 2026, to consider whether behavioural-finance concepts could help explain investor reactions.

OBJECTIVES OF THE STUDY

The main objectives of this study are:

1. To analyse HAL's daily closing share price over the selected period.
2. To calculate daily returns and identify the five largest positive and negative returns.
3. To examine trading volume during days with unusually large price movements.
4. To identify major company announcements around those dates.
5. To compare HAL's returns with the NIFTY 50 index.
6. To evaluate possible behavioural explanations for observed price movements.
7. To examine the market reaction around an official company announcement.

DATA AND METHODOLOGY

Data Sources

The study uses the following data:

- Daily historical share-price and trading-volume data for HAL obtained from the NSE website.
- Historical NIFTY 50 index data obtained from the NSE website.
- HAL's official exchange filing concerning the AMCA programme, dated 4 February 2026.

The HAL dataset covers 14 October 2025 to 14 October 2026. The NIFTY 50 data were collected for 3 February to 6 February 2026 to compare market movements around the selected announcement.

Calculation of Daily Returns

Daily returns measure the percentage change in a share's closing price compared with the previous trading day's closing price.

The formula used is:

Daily Return (%) = [(Today's Closing Price − Previous Trading Day's Closing Price) / Previous Trading Day's Closing Price] × 100

A positive return indicates that the closing price increased, while a negative return indicates that it decreased.

Method of Analysis

Microsoft Excel was used to organise the data, calculate daily returns, identify the largest positive and negative movements, examine trading volumes and prepare charts. HAL's returns were then compared with NIFTY 50 returns for selected dates.

The results are descriptive. They help identify patterns but cannot, by themselves, establish the exact cause of a share-price movement.

ANALYSIS OF HAL'S DAILY CLOSING PRICE

HAL's daily closing prices were plotted over the selected period using a line chart. The chart shows how the share price changed over time and helps identify periods of relatively stable trading and sharp movements.



Figure 1: Daily Closing Price of Hindustan Aeronautics Limited (HAL)

The closing-price chart provides an overview of HAL's price behaviour during the observation period. Changes in the share price may reflect company announcements, investor expectations, sector developments, market conditions and changes in demand and supply.

Sharp price changes are particularly relevant to behavioural finance because they may be associated with changes in investor sentiment. However, a price movement alone does not prove that investors acted irrationally.

ANALYSIS OF THE LARGEST POSITIVE AND NEGATIVE RETURNS

Daily returns were calculated to identify the five largest positive and five largest negative movements in HAL's share price.

The following values are taken from the current Excel analysis and should be checked against the original NSE data before final submission.

Five Largest Positive Returns

category	Date	Daily return(%)	Trading Volume
Largest Positive	28-01-2026	6.34	1837685
Largest Positive	06-08-2026	5.92	4796689
Largest Positive	01-04-2026	5.26	2246768
Largest Positive	04-05-2026	5.09	2841170
Largest Positive	17-06-2026	4.81	1421845

The positive returns indicate days when HAL's closing price increased considerably compared with the previous trading day. Such movements may occur due to favourable company news, positive expectations about future earnings, new orders, sector developments or changes in investor sentiment.

Five Largest Negative Returns

Largest negetive	04-02-2026	-5.67	5290920
Largest negetive	01-02-2026	-5.25	4034039
Largest negetive	15-05-2026	-4.81	2201976
Largest negetive	05-02-2026	-4.31	6343724
Largest negetive	15-09-2026	-4.28	809202

. **1 February 2026 was a Sunday**, so it should not normally appear as an NSE trading date. Check the original source data and correct the date and corresponding return before submitting the final report.

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Figure 2: Largest Positive and Negative Daily Returns of HAL

Interpretation

The results show that HAL experienced both substantial positive and negative daily returns during the selected period. Large movements may indicate that investors were responding to new information or revising their expectations.

Trading volume can provide additional information about market activity. For example, high volume alongside a large price decline may indicate increased selling activity or disagreement among investors.

However, high volume does not necessarily indicate panic selling or irrational behaviour. It can also result from institutional trading, changes in portfolio positions or the arrival of important information.

COMPARISON OF HAL WITH NIFTY 50

The NIFTY 50 index was used as a benchmark to assess whether HAL's price movements were consistent with the broader Indian equity market.

The following table compares daily returns for 4, 5 and 6 February 2026.

Date	HAL Return (%)	NIFTY 50 Return (%)
04-Feb-26	-5.67	0.19
05-Feb-26	-4.31	-0.52
06-Feb-26	0.79	0.20

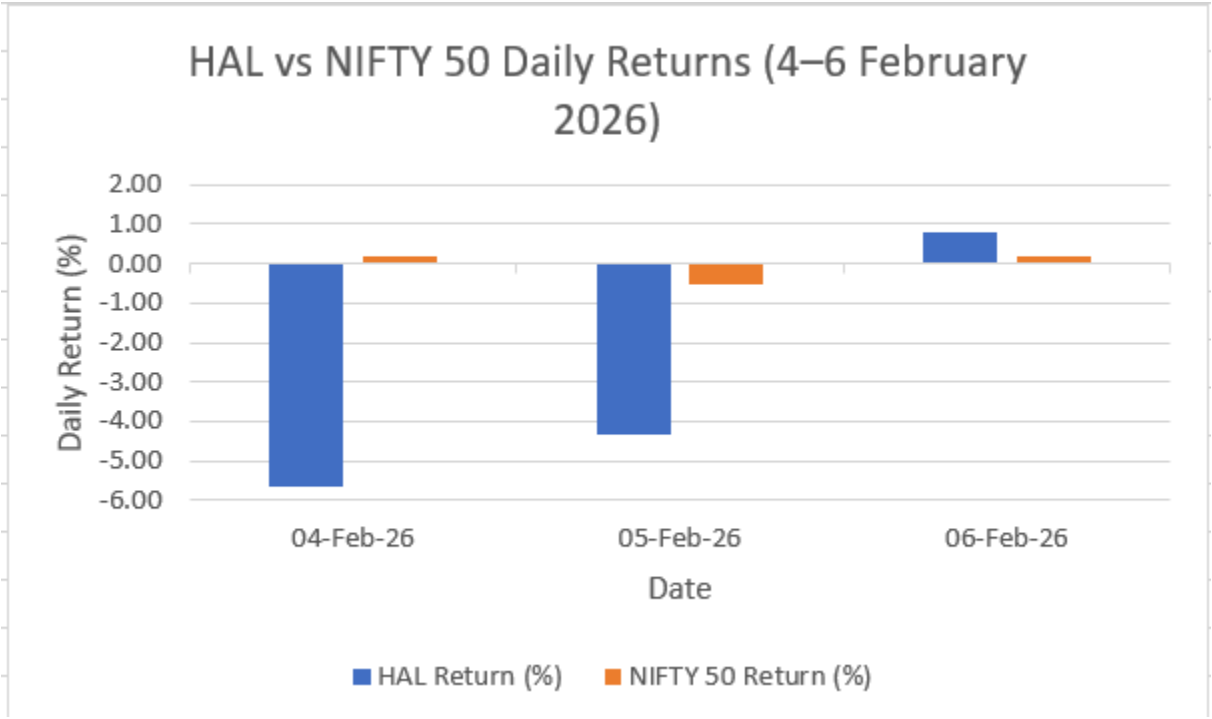


Figure 3: Comparison of HAL and NIFTY 50 Daily Returns, 4–6 February 2026

Interpretation of the Comparison

4 February 2026: HAL's share price fell by approximately 5.67%, while the NIFTY 50 rose by approximately 0.19%. This indicates that HAL moved in the opposite direction to the broader market on that day.

5 February 2026: HAL declined by approximately 4.31%, while the NIFTY 50 fell by approximately 0.52%. HAL's decline was considerably larger than the market's decline.

6 February 2026: HAL recovered by approximately 0.79%, compared with an increase of approximately 0.20% in the NIFTY 50.

Overall, HAL's price movements were considerably larger than the movements in the broader market over these three sessions. This suggests that general market conditions alone may not explain HAL's movements. Company-specific information, investor expectations or other factors may also have played a role.

Nevertheless, a comparison covering only three sessions is limited. It cannot establish the exact cause of the observed movements.

ANALYSIS OF THE OFFICIAL COMPANY ANNOUNCEMENT

Announcement Selected

The official announcement selected for this study is HAL's media release titled:

"Clarification regarding AMCA Programme"

Announcement date: 4 February 2026
Company: Hindustan Aeronautics Limited
Source: Official HAL filing available through the NSE website.

Official announcement PDF:

https://nsearchives.nseindia.com/corporate/HAL_04022026161746_Media_Release_04022026.pdf

What Did the Company Announce?

HAL issued a clarification in response to media reports concerning the Advanced Medium Combat Aircraft (AMCA) programme.

The company stated that it had not received official communication regarding the reports and was therefore not in a position to comment on them at that stage.

HAL also reiterated that it had a confirmed order book and a production and execution pipeline extending to 2032. The company referred to other strategic programmes, including the Indian Multi Role Helicopter (IMRH), LCA Mk2 and Combat Air Teaming System (CATS).

The announcement also discussed HAL's diversification into civil aviation through platforms such as Dhruv NG, Hindustan 228 and SJ 100.

The release therefore combined a clarification about the AMCA reports with a reaffirmation of HAL's order book, ongoing programmes and longer-term growth prospects.

Question 1: What information did the company announce?

HAL clarified that it had not received official communication regarding the reported AMCA programme developments and could not comment on the reports at that stage.

The company reiterated its confirmed order book, production pipeline and other strategic programmes. The statement also highlighted its longer-term growth prospects and civil-aviation initiatives.

Question 2: Did the share price rise or fall around the announcement?

HAL's share price fell by approximately 5.67% on 4 February 2026 and 4.31% on 5 February 2026. It then recovered by approximately 0.79% on 6 February.

An important timing detail must be considered: the announcement was published on the NSE at approximately 4:17 PM on 4 February, after regular market trading hours.

Therefore, the announcement cannot be treated as the cause of HAL's decline during the 4 February trading session. The price movement on 5 February is more relevant when examining a possible market reaction after the announcement.

However, other news and market developments may also have influenced the share price.

Question 3: Did trading volume change?

The Excel data show that HAL's trading volume was approximately 5.29 million shares on 4 February and 6.34 million shares on 5 February.

Trading volume therefore increased between these two sessions, while HAL experienced another sharp price decline on 5 February.

This indicates substantial trading activity during the period. However, to establish whether the volume was unusually high, it should be compared with HAL's average or median daily volume over a longer period.

Volume alone cannot establish whether investors were reacting emotionally or trading on new information.

Question 4: Was the price movement different from the broader market?

Yes. HAL's price movements were considerably larger than those of the NIFTY 50.

On 4 February, HAL declined by approximately 5.67%, whereas the NIFTY 50 increased by 0.19%.

On 5 February, HAL declined by approximately 4.31%, while the NIFTY 50 declined by only 0.52%.

On 6 February, HAL recovered by approximately 0.79%, compared with an increase of 0.20% in the NIFTY 50.

This comparison suggests that the broader market alone does not explain HAL's movements during this period. Company-specific information or changes in investor expectations may also have contributed.

Question 5: Could overreaction, underreaction or herding behaviour be plausible explanations?

Several behavioural-finance explanations are possible.

Overreaction: Investors may have reacted strongly to uncertainty surrounding the AMCA programme, particularly if they interpreted media reports negatively before receiving complete information. Such a reaction could contribute to a sharp price decline. However, the available data do not prove that overreaction occurred.

Herding behaviour: Some investors may have followed the actions of other market participants, increasing selling pressure if negative sentiment spread. This is a possible explanation, but it cannot be confirmed from price and volume data alone.

Underreaction: Underreaction occurs when investors adjust prices gradually as information becomes available. If the market incorporated information over several trading sessions rather than immediately, underreaction could be considered. The short observation window used here is insufficient to establish this pattern.

These explanations should therefore be treated as hypotheses rather than proven causes.

Question 6: What other explanations should be considered?

Other explanations include:

1. **Company-specific developments:** Investors may have responded to news or expectations concerning defence programmes, aircraft deliveries, production schedules or future orders.
2. **Profit-taking:** Some investors may have sold shares to realise profits after earlier price increases.
3. **Sector developments:** News affecting the defence and aerospace sector may have influenced investor expectations about HAL.
4. **Institutional trading:** Changes in institutional positions or portfolio allocation may have affected buying and selling pressure.
5. **Market liquidity and uncertainty:** Changes in trading activity, liquidity or investor risk appetite may have contributed to the price movements.
6. **Broader economic conditions:** Interest rates, macroeconomic developments or other market-wide factors may also have affected investor decisions.

The timing of relevant company announcements should be checked before attributing a particular price movement to any one event.

BEHAVIOURAL FINANCE INTERPRETATION

The observed price movements provide a useful case for considering how investor psychology may influence financial markets.

HAL's sharp decline on 4 February, followed by another substantial decline on 5 February, suggests that the stock experienced significant selling pressure during this period. The divergence between HAL and NIFTY 50 indicates that the decline was not simply a reflection of the broader market.

One possible behavioural explanation is **investor overreaction**. Uncertainty surrounding a major defence programme may have caused some investors to revise their expectations rapidly. If investors placed excessive weight on negative reports, this could have amplified the price decline.

Another possible explanation is **herding behaviour**, in which investors follow the actions of other market participants rather than independently evaluating available information. Such behaviour can potentially intensify price movements.

However, behavioural explanations are not the only possibilities. Investors may have been responding rationally to changing expectations about future business prospects, programme participation or other company-specific information.

Furthermore, the timing of the official AMCA clarification is important. Since the announcement was published after the market closed on 4 February, it cannot explain the price decline that had already occurred during that session.

Consequently, the data suggest that company-specific factors may have influenced HAL's price movements, but they do not establish that behavioural biases caused the decline.

ALTERNATIVE EXPLANATIONS

At least two alternative explanations should be considered before concluding that behavioural biases caused HAL's price movements.

Alternative Explanation 1: Changes in Company Fundamentals and Expectations

The share price may have declined because investors revised their expectations about HAL's future revenue, production schedules, programme participation or execution risks. Such a response could reflect rational reassessment of expected future cash flows rather than irrational behaviour.

Alternative Explanation 2: Profit-Taking and Portfolio Adjustments

Investors may have sold shares to realise profits, reduce exposure or rebalance their portfolios. Institutional investors may also have adjusted their positions for reasons unrelated to the AMCA clarification.

Alternative Explanation 3: Defence-Sector News and Broader Conditions

Developments affecting the defence sector, changes in market sentiment, liquidity conditions or other economic factors may have influenced HAL's price independently of the selected announcement.

These alternatives demonstrate why price movements should not automatically be attributed to investor psychology.

LIMITATIONS OF THE STUDY

This study has several limitations.

First, the comparison with NIFTY 50 focuses on only three trading sessions. A longer event window would provide a more complete understanding of the price movement.

Second, price and volume data cannot directly reveal investors' intentions or psychological states. Therefore, overreaction, underreaction and herding cannot be confirmed from this analysis alone.

Third, other company announcements and sector developments may have influenced HAL's price during the selected period.

Finally, the date listed as 1 February 2026 in the extreme-return table requires verification against the original NSE data because it was a Sunday. The date and corresponding return must be corrected before the report is submitted.

CONCLUSION

This report examined HAL's daily closing prices, daily returns, trading volumes and price movements around an official company announcement. HAL's returns were also compared with the NIFTY 50 index to assess whether its performance differed from the broader market.

The analysis showed that HAL declined by approximately 5.67% on 4 February 2026 and 4.31% on 5 February, while the NIFTY 50 rose slightly on the first date and declined by only 0.52% on the second. HAL subsequently recovered by approximately 0.79% on 6 February.

These differences suggest that general market conditions alone may not explain HAL's price movements during the selected period. Company-specific developments and changes in investor expectations may have played a role.

Behavioural-finance concepts such as overreaction and herding provide possible explanations for the observed pattern. Nevertheless, the available evidence does not establish that these behaviours caused the decline. Other explanations, including changes in expectations, profit-taking and institutional trading, must also be considered.

An important finding is that HAL's AMCA clarification was published after market hours on 4 February. Therefore, it cannot explain the decline that occurred during that day's trading session.

Overall, this case illustrates the importance of combining price data, trading volume, market benchmarks and announcement timing when studying investor behaviour. It also demonstrates why behavioural explanations should be evaluated alongside fundamental and market-based explanations.

REFERENCES

1. **National Stock Exchange of India (NSE).** HAL historical security-wise data for 14 October 2025 to 14 October 2026. Data downloaded for academic analysis.
2. **National Stock Exchange of India (NSE).** Historical NIFTY 50 index data for 3–6 February 2026.
3. **Hindustan Aeronautics Limited (HAL).** “Clarification regarding AMCA Programme,” 4 February 2026. Official exchange filing.

Official announcement:

https://nsearchives.nseindia.com/corporate/HAL_04022026161746_Media_Release_04022026.pdf